

A KERNEL BACKGROUND PURIFICATION BASED ANOMALY TARGET DETECTION ALGORITHM FOR HYPERSPECTRAL IMAGERY

Yan Zhang¹, Mingming Xu¹, Yanguo Fan¹, Yuxiang Zhang² and Yanni Dong²

1. School of Geosciences, China University of Petroleum (East China), P.R. China.
2. Institute of Geophysics and Geomatics, China University of Geosciences, P.R. China.

ABSTRACT

In traditional anomaly detection algorithms, the background information is approximately described by whole hyperspectral imagery. However, the disparity between true and estimated background information would influence the performance of detection algorithms using background information. Considering this problem, a kernel background purification based anomaly target detection method is proposed in this paper. The main idea of the proposed method is to estimate background information more accurately. It contains two main steps: Firstly, the pure background pixel set extraction using the kernel-based method. Secondly, background covariance matrix estimation by extracted pure background pixel set. Experiments implemented on San Diego and PHI data indicate that the proposed method performed better than global Reed-Xiaoli detector (RXD), kernel RXD, and collaborative representation detector (CRD).

Index Terms— anomaly detection, hyperspectral imagery, background purification, kernel method, spectral angle matching

1. INTRODUCTION

Hyperspectral imaging technology develops rapidly with high spectral resolution in recent years. High spectral resolution means that the reflectance characteristic of each ground object will be more detailed, which gives great potential to distinguish different ground objects using inherent spectral reflectance characteristics [1]. Hyperspectral imagery (HSI) has been developed a widely range of applications, such as classification and target detection. Latter application is divided into target detection with prior target knowledge and anomaly target detection without prior target knowledge. In many actual cases, the prior target knowledge is difficult to acquire, and therefore anomaly target detection algorithms are needed [2].

RXD [3] is one of the most well-known benchmark anomaly detection algorithms. Many improved algorithms based on RXD are proposed, such as local RXD, subspace RXD [4], and kernel RXD [5]. In kernel RXD and RXD, the background information is estimated by the whole or local HSI, which will lead to the disparity of estimated and true background information. Actually, the accurate background information estimation is important for background-modeling anomaly target detection algorithms. Kernel-based method is also used

in the proposed algorithm. Kernel-based method is a nonlinear method in which the separability of ground objects can be magnified by mapping input data into high-dimensional feature space [6]. Some kernel-based anomaly detection algorithms perform well, such as kernel collaborative representation detector [7], cluster kernel RXD [8], sparse kernel-based hyperspectral anomaly detection [9]. In this paper, we proposed a background purification based RXD (KBPB-RXD) algorithm to eliminate the effect of anomaly pixels in estimating the background information and it is designed as a global target detector.

2. PROPOSED METHOD

2.1 RXD algorithm

For HSI $X \in R^{B \times N}$ (B is the number of spectral bands and N is the total number of pixels), each pixel can be labeled as a vector $\{x_i\}_{i=1}^N$. The global RXD detector is defined as:

$$D_{RXD} = (x - \mu_0)^T \Sigma^{-1} (x - \mu_0) \quad (1)$$

where μ_0 is the mean value vector of overall pixels; Σ is the background covariance matrix estimated from overall pixels.

2.2 kernel RXD algorithm

In kernel RXD, the kernel-based method is used to improve the detection performance by mapping original data into high-dimensional space. The kernel function is utilized to build the relationship of original space and high dimensional feature space. One widely used kernel function is the Gaussian radial basis function (RBF) [10], the formula expression of RBF is:

$$k(x_i, x_j) = \exp(-\|x_i - x_j\|^2 / c) \quad (2)$$

where c is a non-negative constant.

Subsequently, kernel RXD algorithm can be represented as:

$$D_{KRXD}(\Phi(x)) = (\Phi(x) - \hat{\mu}_\Phi)^T \hat{C}_\Phi^{-1} (\Phi(x) - \hat{\mu}_\Phi) \quad (3)$$

where $\Phi(x)$ is the mapped data in high dimensional space. $\hat{\mu}_\Phi$ and $\hat{C}_\Phi$ are corresponding mean value and background covariance matrix which are estimated in

feature space.

2.3 kernel background purification based RXD algorithm

2.3.1 kernel spectral angle distance computation

Analyzing the characteristics of anomaly target[11], we can draw conclusions that: 1) global anomaly target pixels are located far from the center of probability distribution; 2) background pixels are adjacent with the center of probability distribution. The distance between the center of probability distribution and the pixel under test can be as a criterion to remove the outliers in the process of background purification. The spectral angle distance is chosen as the distance measurement in this paper. To further improve the performance of KBPB-RXD, principal component analysis (PCA) will be used to remove the noise firstly.

The spectral angle distance between the center of probability distribution and pixel x_i is expressed as:

$$S_i(x_i, \mu) = \arccos\left(\frac{\langle x_i, \mu \rangle}{\|x_i\| \|\mu\|}\right) \quad (4)$$

where μ is the mean value of HSI. In Eq.(4), we introduce the kernel-based method into the proposed method. The kernel spectral angle distance is expressed as:

$$\Phi S_i(x_i, \mu) = \arccos\left(\frac{\langle \Phi(x_i), \hat{\mu}_\Phi \rangle}{\|\Phi(x_i)\| \|\hat{\mu}_\Phi\|}\right) \quad (5)$$

In Eq.(5),

$$\begin{aligned} \|\Phi(x_i)\| &= \sqrt{\langle \Phi(x_i), \Phi(x_i) \rangle} = \sqrt{\exp(0)} = 1 \\ \|\hat{\mu}_\Phi\| &= \sqrt{\langle \hat{\mu}_\Phi, \hat{\mu}_\Phi \rangle} = \sqrt{\exp(0)} = 1 \\ \langle \Phi(x_i), \hat{\mu}_\Phi \rangle &= \exp((- \|x_i - \mu\|^2) / c) \end{aligned} \quad (6)$$

Then, Eq.(5) can be simplified as:

$$\Phi S_i(x_i, \mu) = \arccos(\exp((- \|x_i - \mu\|^2) / c)) \quad (7)$$

2.3.2 background purification

After kernel spectral angle distance computation, the kernel spectral angle distances ΦS_i can be obtained. The quartile method which is more robust for the outliers is introduced to implement the process of removing the outlier using the kernel spectral angle distances. The interquartile range is defined as follow:

$$dF = F_U - F_L \quad (8)$$

where F_U is upper quartile which locates at 25% of the spectral angle distances from small to large; F_L is lower quartile which locates at 75% of the spectral angle distances from small to large.

The threshold which is used to remove the outlier is calculated by:

$$threshold_{\max} = M + \alpha \cdot dF \quad (9)$$

where M is median value of the spectral angle distances;

α is adjustable constant, and its value can be 0.5, 1.0, 2.0 and so on.

The mean value $Mean_{ii=1}^k$ of kernel spectral angle distances is updating among repeating the process of removing outliers. And k is the maximum number of iterations.

According to aforementioned content, the kernel spectral angle value between background pixel and mean value is relatively small. The relationship between $Mean_i$ and $Mean_{i-1}$ can be depicted as follow in the process of removing of outliers:

$$Mean_i < Mean_{i-1} \quad (10)$$

Kernel spectral angle is non-negative value:

$$Mean_i = \sum_{i=1}^n \Phi S_i / n > 0 \quad (11)$$

Analyzing the sequence $Mean_{ii=1}^k$, we can draw a conclusion:

$$\lim_{i \rightarrow k} (Mean_{i-1} - Mean_i) = 0 \quad (12)$$

If $Mean_{i-1} - Mean_i$ is less than a threshold value, the iteration of background purification would be stopped.

2.3.3 KBPB-RXD algorithm

Pure background pixels set is selected by kernel background purification processing. KBPB-RXD is defined as:

$$D_{KBPB-RXD}(x) = (x - \mu_0)^T \tilde{\Sigma}^{-1} (x - \mu_0) \quad (13)$$

where $\tilde{\Sigma}$ is re-estimated background covariance matrix by pure background pixels set after kernel background purification processing.

3. EXPERIMENTAL RESULTS

3.1 hyperspectral data

Two data set are used to validate the algorithm performance. San Diego airport data was collected by AVIRIS sensor with a size of 100×100 pixels which is depicted in Fig.1(a). Three aircraft were selected as anomaly target under test in the scene. The ground-truth location of anomaly target is depicted in Fig.1(b).



Fig.1. (a) AVIRIS image of the San Diego airport scene. (b) Ground-truth map of 58 anomalous pixels.

PHI hyperspectral image is a simulated data with a

size of 240×240 pixels which is depicted in Fig.2(a). 100 anomaly pixels are included in the scene. The tenth line of anomaly target is single-pixel. Pixels in ninth line are composed of ten percent background and ninety percent anomaly target. Pixels in eighth line are composed of twenty percent background and eighty percent target. Remained anomaly pixels obey the same rule.



Fig.2. (a) The gray image of PHI image. (b) Ground-truth map of 100 anomalous pixels.

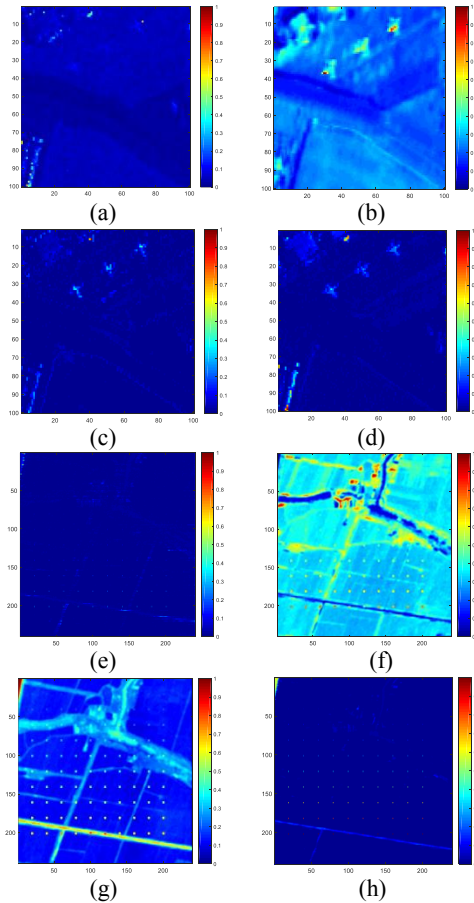


Fig.3 Anomaly detection results of (a)global RXD, (b)KRXD, (c)CRD, (d)KBPB-RXD implemented on San Diego data, (e)global RXD, (f)KRXD, (g)CRD, (h)KBPB-RXD implemented on PHI data.

3.2 experimental results

In Section 3.2, the experimental results implemented on two data set were discussed. The experimental results are

shown in Fig.3. The evaluating indicator receiver operating characteristic curve (ROC) are depicted in Fig.4. The size of outer and inner windows in Fig.3(c) are 5×5 and 3×3 , in Fig.3(g) are 3×3 and 1×1 . Adjustable constant in Fig.3(d) is 1.5, in Fig.3(h) is 0.5. The threshold value in Fig.3(d) and Fig.3(h) are both set 0.0001.

From the experiment results, we can see that the KBPB-RXD performed better than another three methods. Anomaly target and background can be separated more distinct using the KBPB-RXD algorithm. It also has great performance in dealing with sub-pixel anomaly target detection. Global RXD, KRXD and CRD are affected by deviation of background information estimation, and the ability of separating background and anomaly target is weakened.

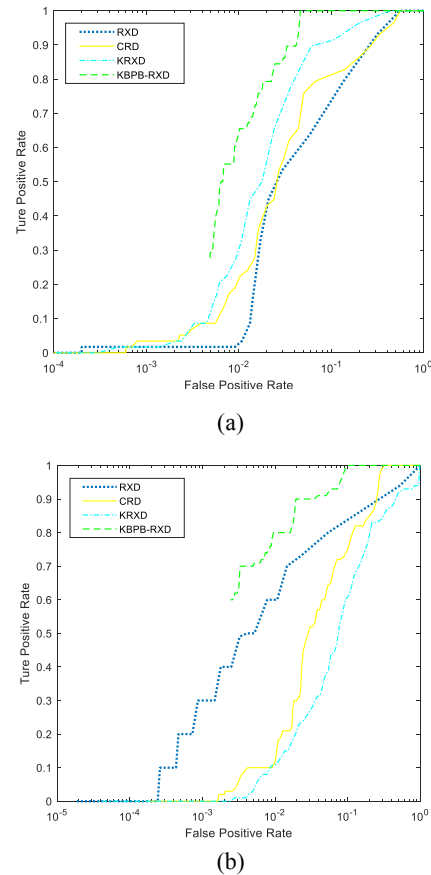


Fig.4 ROC curve of (a)San Diego data, (b)PHI data.

4.CONCLUSION

In this paper, a background purification-based anomaly target detection method is proposed. It applies the kernel-based method before estimating the background covariance matrix. The background covariance matrix can be estimated more precise by removing the suspected anomaly target pixels from background pixel set. In this paper, two data set are used to validate the proposed method. Experimental results indicate that the proposed method can improve detection performance effectively.

The background purification processing can also be used to improve other anomaly target detection methods which involve background information estimation.

5. ACKNOWLEDGMENTS

This study was supported in part by the National Natural Science Foundation of China under grant 61701542, in part by the Shandong Provincial Natural Science Foundation of China under grant ZR2017BF038, in part by the Fundamental Research Funds for the Central Universities under grant 17CX02002A.

6. REFERENCES

- [1] Q. Tong, B. Zhang and L. Zhang, " Current progress of hyperspectral remote sensing in China," *Journal of Remote Sensing*, vol. 20, no.5, pp. 689-707, 2016.
- [2] L. Zhang, L. Zhang, D. Tao and B. Du, " Hyperspectral Remote Sensing Image Subpixel Target Detection Based on Supervised Metric Learning," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 52, no. 8, pp. 4955-4965, 2014.
- [3] I. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Transactions on Acoustics, Speech and Signal Processing*, vol. 38, no.10, pp. 1760-1770, 1990.
- [4] E. Lo, "Maximized subspace model for hyperspectral anomaly detection," *Pattern Analysis and Applications*, vol. 15, no.3, pp. 225-235, 2012.
- [5] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: a nonlinear anomaly detector for hyperspectral imagery," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 43, no.2, pp. 178-194, 2005.
- [6] T. Hofmann, B. Schölkopf and A. J. Smola, " Kernel Methods in Machine Learning," *Annals of Statistics*, vol.36, no.3, pp. 1171-1220, 2008.
- [7] W. Li and Q. Du, "Collaborative Representation for Hyperspectral Anomaly Detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 53, no.3, pp. 1463-1474, 2015.
- [8] Z. Jin, C. Kwan, B. Ayhan and M.T. Eismann, " A Novel Cluster Kernel RX Algorithm for Anomaly and Change Detection Using Hyperspectral Images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 54, no.11, pp. 6497-6504, 2016.
- [9] P. Gurram, H. Kwon and T. Han, " Sparse Kernel-Based Hyperspectral Anomaly Detection," *IEEE Geoscience and Remote Sensing Letters*, vol. 9, no.5, pp. 943-947, 2012.
- [10] U. Ravale, N. Marathe and P. Padiya, "Feature Selection Based Hybrid Anomaly Intrusion Detection System Using K Means and RBF Kernel Function," *Procedia Computer Science*, vol. 45, no.39, pp. 428-435, 2015.
- [11] X. Geng and Y. Zhao, " Principle of small target detection for hyperspectral imagery," *Science in China (Series D)* , vol. 50, no.8, pp. 1225-1231, 2007.